DIPLOMA IN ELECTRONICS / ELECTRONICS AND COMMUNICATION / BIOMEDICAL ENGINEERING • SEMESTER 4

Microcontroller and Applications

This corrected edition makes 4041 a proper 8051 engineering handbook. It explains architecture, memory map, PSW, assembly programming, addressing modes, timers, interrupts, serial communication and interfacing with motors, keyboard, LCD, ADC, DAC and LM35 using diagrams, code patterns, solved problems and field notes.

COURSE CODE	CREDITS
4041	4

PERIODS	TYPE
60	Program Core

8051 □□□□□□□□□□□□□ pin names □□□□□□□ □□□□□□□□. Memory, SFR, PSW flags, instruction effect, timer calculation, interrupt priority, port interfacing □□□□□□□ □□□□□□□□□□ □□□□□□□□□□□□□□□.

COURSE BLUEPRINT

What this handbook is expected to teach

This section converts the dull syllabus table into a usable engineering route: prerequisite knowledge, outcome target, field meaning, diagrams to draw, and problems to solve.

Official objectives

- ✓ Equip students with 8051 architecture, interfacing and ALP knowledge.
- ✓ Develop simple interfacing applications for daily use.
- ✓ Prepare students for troubleshooting microcontroller-based equipment.
- ✓ Connect programs with real input and output devices.

Prerequisite stack

- ✓ Basic engineering mathematics and number systems.
- ✓ Digital electronics, codes, combinational and sequential circuits.
- ✓ Memory concepts and logic gates.
- ✓ Basic electronics and interfacing ideas.

How to use

1. Read the concept explanation.
2. Study the diagram or waveform.
3. Solve the worked example without looking.
4. Write the expected-answer format.

CO	Outcome	Hours	Engineering meaning
CO 1	Explain features and block diagram of 8051.	12	Understand CPU, memory, ports, SFRs and PSW.
CO 2	Develop simple ALP for 8051.	17	Write programs for arithmetic, BCD/ASCII conversion, delay and port operation.
CO 3	Explain interrupts, timers and serial communication.	14	Configure IE, IP, TMOD, TCON, SCON, SBUF and PCON.
CO 4	Illustrate interfacing of I/O devices with 8051.	15	Connect motors, keyboard, LCD, ADC, DAC and LM35 with correct signals.

REDESIGNED SYLLABUS STRUCTURE

Module map with diagrams, applications and problem targets

The old plain syllabus table is not enough. This map tells what to draw, what to explain, where the marks come from, and how the concept is used in industry.

Module	Core topics	Must draw / visualize	Must solve / apply	Hours
M1: Features, block diagram and memory organization of 8051	Microprocessor vs microcontroller, features, 8051 block diagram, data memory, program memory and PSW.	8051 block diagram, program memory map, data memory map, PSW bit layout.	Memory location interpretation and PSW flag reasoning.	12
M2: Assembly language programming, addressing modes and instructions	ALP structure, addressing modes, data transfer, arithmetic, logical, compare, rotate, swap, jump, loop, call, delay, bit instructions and simple programs.	Program structure, addressing mode examples, flowchart for delay and conversion programs.	8-bit addition, subtraction, multiplication, division, ASCII/BCD/HEX conversion and port I/O.	17
M3: Interrupts, timers and serial communication in 8051	Interrupt concept, ISR, IE, IP, TMOD, TCON, timer modes, delay calculation, SCON, SBUF, serial modes and PCON.	Interrupt priority flow, timer mode table, serial communication block, RS232 connection concept.	Timer initial value, baud rate concept and interrupt priority questions.	14
M4: Interfacing I/O devices with 8051	Stepper motor, DC motor speed control, 4x4 keyboard, 16x2 LCD, ADC, DAC and LM35 temperature sensor interfacing.	Stepper motor driver, DC motor driver, matrix keyboard, LCD pin block, ADC/DAC/LM35 interface block.	I/O table, scan algorithm and interfacing sequence questions.	15

Features, block diagram and memory organization of 8051

8051 is not only a chip name. It is a complete small computer: CPU, memory, I/O ports, timers, interrupt system and serial port in one package.

Textbook explanation

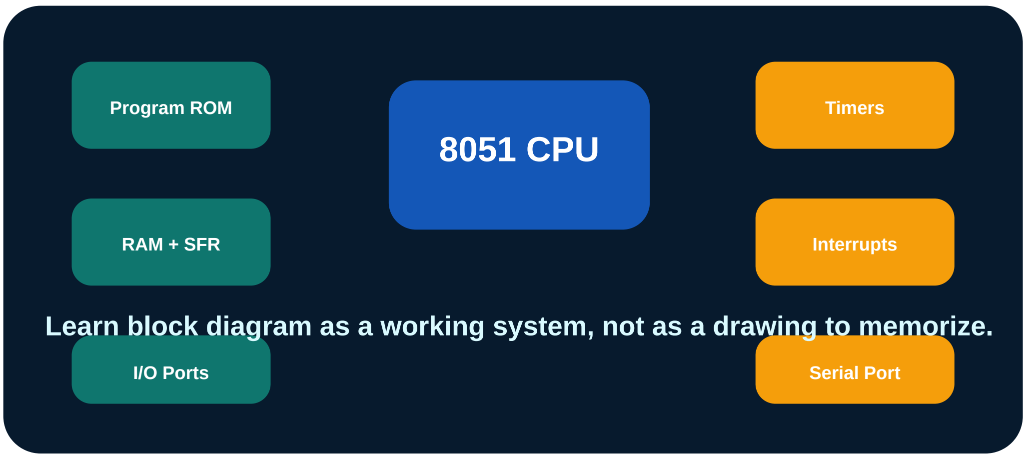
A microprocessor normally needs external memory and I/O devices. A microcontroller includes CPU, memory, ports and peripherals on the same chip.

8051 has 4 KB program ROM in classic version, 128 bytes internal RAM, four 8-bit ports, two timers, serial port and interrupt system.

Internal RAM includes register banks, bit-addressable area and general purpose RAM. SFRs control ports, timers, serial communication and interrupts.

PSW contains flags such as CY, AC, OV and P and register bank selection bits.

> 8051 architecture answer-□ CPU, ROM, RAM, ports, timer, serial port, interrupt, bus □□□□□□□ diagram □□□ □□□□□□□□□□.



8051 internal architecture.

Engineering subtopics

Microprocessor vs microcontroller

System-level difference.

- ✓ Microcontroller has built-in memory and I/O.
- ✓ Microprocessor is more general and often needs external chips.

8051 features

Main internal resources.

- ✓ 8-bit CPU.
- ✓ Four 8-bit ports.
- ✓ Timers and serial port.

Program memory

Stores instructions.

- ✓ Addressed by program counter.
- ✓ Can be internal or external.

Data memory

Stores variables and stack.

- ✓ Register banks.
- ✓ Bit-addressable area.
- ✓ General RAM.

SFRs

Control registers.

- ✓ P0 to P3.
- ✓ TMOD and TCON.
- ✓ SCON and SBUF.

PSW

Program status word.

- ✓ Carry flag.
- ✓ Overflow flag.
- ✓ Parity flag.
- ✓ Register bank selection.

Important formulas / rules

> 8051 has four ports × 8 bits = 32 I/O lines.

> Internal RAM lower area includes register banks, bit addressable area and general RAM.

> PSW flag changes depend on arithmetic or logical instruction result.

Common mistakes

- ✓ Drawing only CPU and forgetting ports.
- ✓ Not separating internal RAM and SFR area.
- ✓ Confusing ROM and RAM.
- ✓ Ignoring PSW flags in ALP questions.

Worked examples inside this module

Why is 8051 called microcontroller?

It includes CPU, memory, I/O ports, timers, serial port and interrupt hardware inside one chip, so it can control devices with fewer external components.

What is PSW used for?

PSW stores status flags and register bank selection bits used by the CPU during program execution.

Assembly language programming, addressing modes and instructions

Assembly programming is exact. Every instruction changes registers, flags, memory or port pins. The correct way to study ALP is to trace each instruction step by step.

Textbook explanation

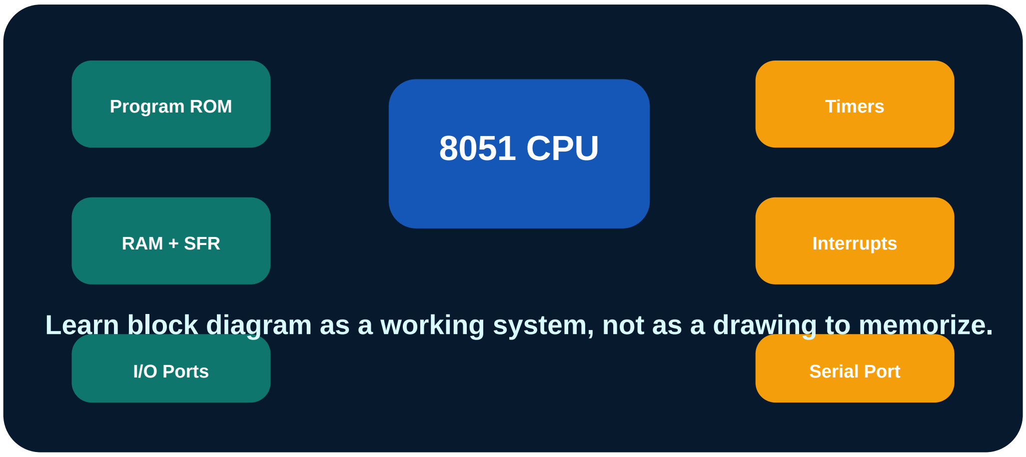
ALP contains labels, mnemonics, operands, comments and directives. A good program also includes clear initialization and end statement.

Addressing modes decide where the operand comes from: immediate, register, direct, indirect and indexed.

Instruction groups include data transfer, arithmetic, logical, compare, rotate, swap, jump, call and bit-level instructions.

Delay programs use loops and machine cycle calculation. Port reading and writing connects software to external hardware.

> ALP answer-□ program □□□□□□ □□□. Algorithm, register use, comments, result location □□□□□□ □□□□□□□□.



Instruction flow inside microcontroller.

Engineering subtopics

ALP structure

Labels, opcode, operands and comments.

- ✓ ORG directive.
- ✓ END directive.
- ✓ Readable comments.

Addressing modes

Operand source method.

- ✓ Immediate.
- ✓ Register.
- ✓ Direct.
- ✓ Indirect.
- ✓ Indexed.

Arithmetic

ADD, SUBB, MUL and DIV.

- ✓ Carry and overflow may change.
- ✓ Result registers must be mentioned.

Logical and rotate

ANL, ORL, XRL, CPL, RL, RR.

- ✓ Used for bit masking.
- ✓ Used for serializing data.

Branching

Jump, loop and call.

- ✓ SJMP, LJMP, DJNZ.
- ✓ Subroutine with CALL and RET.

Port I/O

Reading and writing external pins.

- ✓ Port latch behaviour.
- ✓ Use driver for heavy loads.

Important formulas / rules

> Delay time = machine cycle time × number of executed machine cycles.

> For 12 MHz classic 8051, one machine cycle is about 1 microsecond.

> BCD correction may require decimal adjust after addition.

Common mistakes

- ✓ Writing program without comments.
- ✓ Not clearing carry before subtraction where required.
- ✓ Forgetting result register after MUL or DIV.
- ✓ Using port pin directly for motor current without driver.

Worked examples inside this module

Write logic for 8-bit addition.

Load first number to accumulator, add second number, store result, check carry if required. Example pattern: MOV A,number1; ADD A,number2; MOV result,A.

How is delay generated?

A register is loaded with a count and decremented in a loop using DJNZ. Total delay is calculated from machine cycles used by loop instructions.

Interrupts, timers and serial communication in 8051

Timers, interrupts and serial port make 8051 useful in real machines. Without them, the CPU would waste time waiting for every event manually.

Textbook explanation

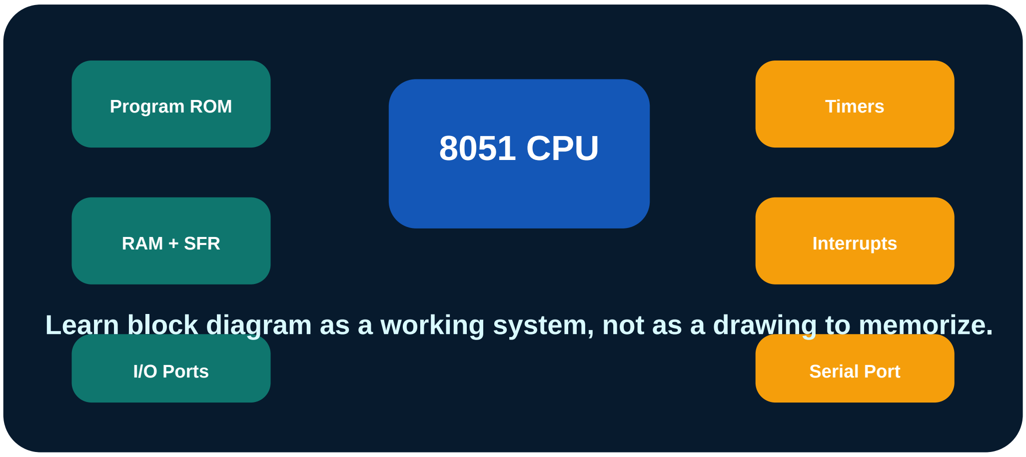
Interrupt allows external or internal event to temporarily stop main program and run an ISR. IE enables interrupts and IP controls priority.

Timers can work in different modes using TMOD. TCON contains timer run and overflow flags.

Timer delay depends on oscillator frequency, machine cycle time, timer mode and initial count.

Serial communication uses SBUF for data and SCON for mode and flags. PCON can affect baud rate.

> Timer calculation-□ oscillator frequency, machine cycle, count, overflow value □□□□□□ clear □□□□□□□□□□.



8051 timers, interrupts and serial port as internal peripherals.

Engineering subtopics

Interrupt concept

Event-driven program execution.

- ✓ Main program is interrupted.
- ✓ ISR executes.
- ✓ RETI returns.

IE and IP

Enable and priority registers.

- ✓ EA global enable.
- ✓ Priority selection.

TMOD

Timer mode register.

- ✓ Timer 0 and Timer 1 modes.
- ✓ Timer or counter selection.

TCON

Run and overflow flags.

- ✓ TR0, TR1.
- ✓ TF0, TF1.

Serial port

UART style communication.

- ✓ SBUF data register.
- ✓ SCON control register.
- ✓ Modes 0 to 3.

Baud rate

Bits per second.

- ✓ Timer often generates baud rate.
- ✓ PCON can double baud in some modes.

Important formulas / rules

Machine cycle frequency = oscillator frequency / 12 for classic 8051.

Timer count required = delay / machine cycle time.

Initial value = maximum count - required count.

Common mistakes

- ✓ Using oscillator period instead of machine cycle time.
- ✓ Forgetting to clear overflow flag.
- ✓ Ending ISR with RET instead of RETI.
- ✓ Confusing SBUF and SCON.

Worked examples inside this module

For 12 MHz 8051, what is one machine cycle?

Classic 8051 divides oscillator by 12. 12 MHz gives 1 MHz machine cycle frequency, so one machine cycle is 1 microsecond.

Why interrupts are useful?

They let CPU respond quickly to important events without continuously polling every input.

Interfacing I/O devices with 8051

Interfacing is where textbook programming becomes real electronics. The microcontroller pin cannot directly drive every load; drivers, pull-ups, ADC and protection matter.

Textbook explanation

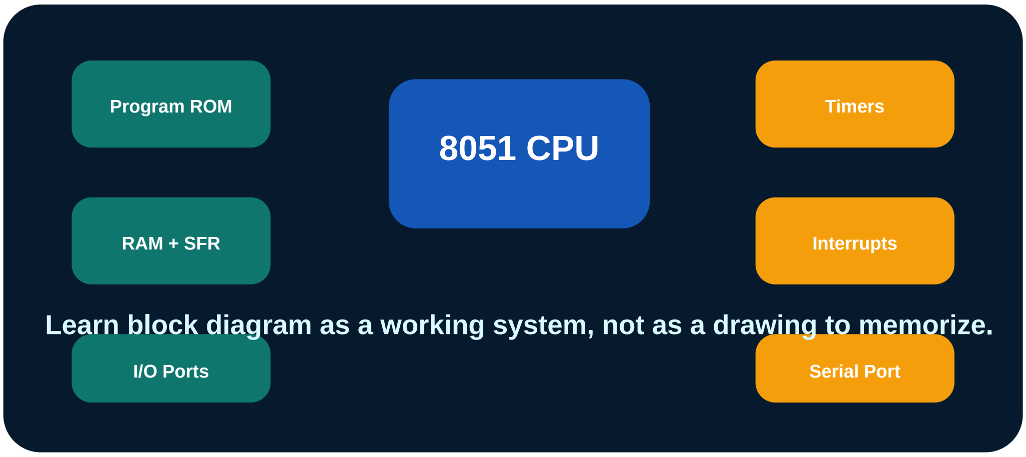
Stepper motor requires a sequence of coil energization. 8051 usually drives it through transistor or driver IC, not directly.

DC motor speed control can use PWM and driver transistor or H-bridge depending on direction requirement.

Matrix keyboard is scanned by driving rows and reading columns. LCD requires command and data sequence with enable signal timing.

Analog sensors such as LM35 require ADC because 8051 is digital. DAC converts digital data to analog output.

> Interfacing diagram-□ driver IC, power supply, common ground, protection diode □□□□□□□ mention □□□□□□□□.



8051 connected to field devices through drivers and converters.

Engineering subtopics

Stepper motor

Moves in steps using coil sequence.

- ✓ Full step and half step.
- ✓ Needs driver stage.

DC motor

Speed control using PWM.

- ✓ Transistor or H-bridge driver.
- ✓ Back EMF protection needed.

Matrix keyboard

Rows and columns scanned by software.

- ✓ Drive one row at a time.
- ✓ Read columns.
- ✓ Debounce needed.

16x2 LCD

Displays characters using command and data pins.

- ✓ RS, RW, EN.
- ✓ 4-bit or 8-bit mode.

ADC and DAC

Connect analog world and digital controller.

- ✓ ADC reads sensors.
- ✓ DAC generates analog voltage.

LM35

Temperature sensor.

- ✓ 10 mV per degree Celsius.
- ✓ Needs ADC input.

Important formulas / rules

LM35 output ≈ 10 mV/°C.

PWM duty cycle controls average motor voltage.

Keyboard scanning = row selection plus column reading.

Common mistakes

- ✓ Connecting motor directly to port pin.
- ✓ Forgetting common ground between 8051 and driver.
- ✓ No debounce for keyboard.
- ✓ Trying to read LM35 without ADC.

Worked examples inside this module

Why driver is required for motor interfacing?

8051 port pin can supply only small current. Motor needs higher current and produces back EMF, so driver transistor or driver IC is required.

How is 4x4 keyboard scanned?

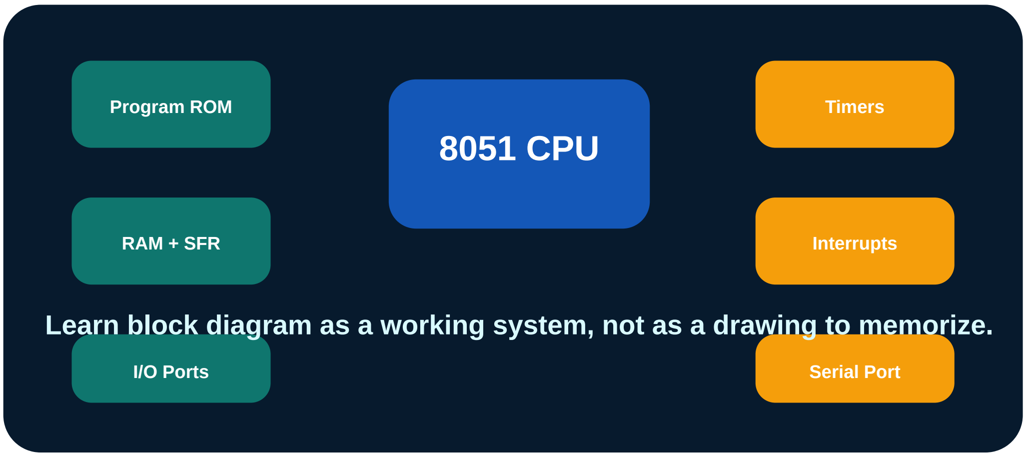
Make one row active at a time and read column lines. The row-column combination identifies the pressed key.

ILLUSTRATION LAB

Images, block diagrams and waveform thinking

For technical subjects, the diagram is half of the answer. These visuals make the lesson look and work like a textbook instead of a plain note.

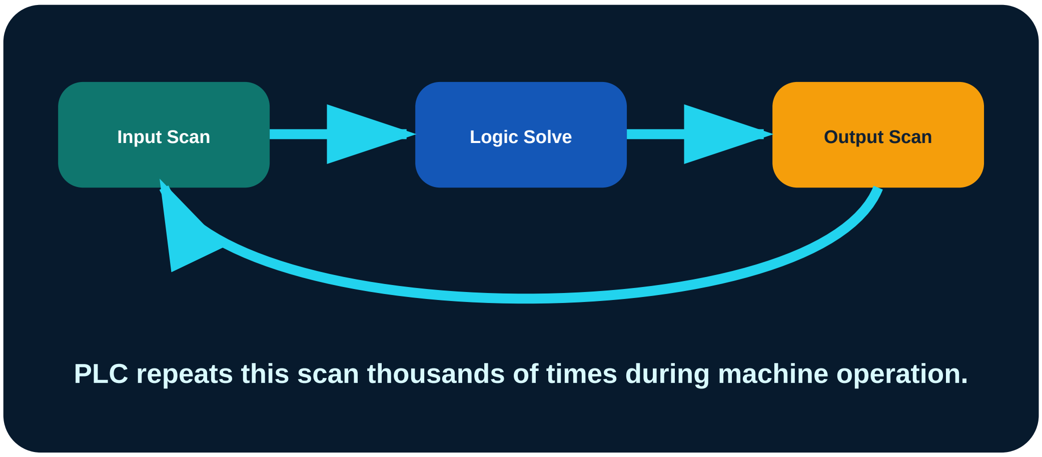
8051 block diagram



CPU, memory, ports, timers, interrupts and serial port.

Use this for architecture answers.

PLC style scan comparison



Microcontroller also repeatedly reads inputs and writes outputs.

Useful for embedded control thinking.

Interfacing chain

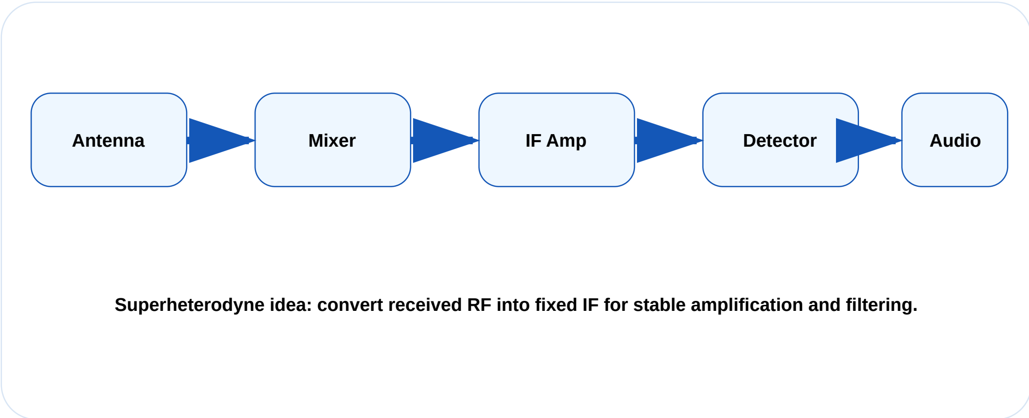
Sensor or keyboard to 8051 to driver to actuator.

Diagram, waveform, block flow and practical meaning should be studied together.

Sensor or keyboard to 8051 to driver to actuator.

Every interface answer needs signal direction.

Serial communication chain



Data is transmitted, received and interpreted through registers.

Useful for SBUF and SCON answers.

PROBLEM LAB

Solved numerical and design problems

Each problem shows the method, not only the answer. This is the difference between a notebook and an engineering handbook.

Machine cycle

Problem: Find machine cycle time for classic 8051 with 12 MHz crystal.

1. Classic 8051 uses 12 oscillator periods per machine cycle.
2. Machine cycle frequency = 12 MHz / 12 = 1 MHz.
3. Time = 1 / 1 MHz.

Answer: 1 microsecond

Timer initial value idea

Problem: Timer mode 1 needs 1000 counts. Find initial value in hex.

1. Mode 1 maximum count is 65536.
2. Initial value = 65536 - 1000 = 64536.
3. Convert 64536 to hex.

Answer: FC18H

LM35 temperature

Problem: LM35 output is 300 mV. Find temperature.

1. LM35 gives 10 mV per degree Celsius.
2. Temperature = 300/10.

Answer: 30 °C

Motor driver reason

Problem: Why is a transistor driver needed for DC motor?

1. 8051 port current is small.
2. Motor requires higher current.
3. Motor back EMF can damage controller.

Answer: Driver provides current gain and protection.

PRACTICE

Expected questions by module

Module 1

- Compare microprocessor and microcontroller.
- List features of 8051.
- Draw and explain 8051 block diagram.
- Explain data memory organization.
- Explain program memory organization.
- Explain PSW register.

Module 2

- Explain structure of ALP.
- Explain addressing modes of 8051.
- Explain data transfer and arithmetic instructions.
- Explain jump, loop and call instructions.
- Develop ALP for addition and subtraction.
- Develop ALP for BCD and ASCII conversion.

Module 3

- Explain interrupt concept, types and ISR.
- Explain IE and IP registers.
- Explain TMOD and timer modes.
- Explain TCON register and delay calculation.
- Explain SCON, SBUF and serial modes.
- Explain baud rate concept.

Module 4

- Explain stepper motor interfacing with 8051.
- Explain DC motor speed control interfacing.
- Explain 4x4 matrix keyboard scanning.
- Explain 16x2 LCD interfacing.
- Explain ADC and DAC interfacing.
- Explain LM35 temperature sensor interfacing.

MODEL PAPER

Full practice question paper and answer guide

This model follows the syllabus order and avoids copying official papers.

COURSE 4041 MICROCONTROLLER AND APPLICATIONS
Time: 3 Hours   Maximum Marks: 100

PART A
1. Define microcontroller.

2. List two features of 8051.
3. What is PSW?
4. Name two addressing modes.
5. What is DJNZ used for?
6. Define ISR.
7. What is TMOD?
8. What is SBUF?
9. Why driver is used for motor?
10. What is LM35 output scale?

PART B

11. Compare microprocessor and microcontroller.
12. Draw 8051 block diagram.
13. Explain 8051 memory organization.
14. Explain addressing modes.
15. Write ALP algorithm for 8-bit addition.
16. Explain interrupts and IE register.
17. Explain timer delay calculation.
18. Explain stepper motor interfacing.

PART C

19. Explain 8051 architecture and PSW.
20. Develop ALP for arithmetic and conversion programs.
21. Explain jump, loop, call and bit instructions.
22. Explain timer registers and serial communication registers.
23. Explain 4x4 keyboard and LCD interfacing.
24. Explain ADC, DAC and LM35 interfacing.
25. Solve timer and LM35 numerical problems.
26. Design an 8051 based temperature display block.

Answer writing guide

- ✓ For architecture answers draw block diagram and explain every block.
- ✓ For ALP questions include algorithm, program and result location.
- ✓ For timer problems show machine cycle calculation.
- ✓ For interfacing questions include driver, signal direction and protection.

REFERENCES

Books and resources

Type	Reference
T1	Muhammad Ali Mazidi and Janice Gillispie Mazidi, The 8051 Microcontroller and Embedded Systems
T2	Subratha Ghoshal, 8051 Microcontroller Internals, Instructions, Programming and Interfacing
T3	Kenneth J Ayala, The 8051 Microcontroller
Online	Electronicshub, Elprocus, TutorialsPoint and 8051 reference resources

Final revision rule: Prepare one diagram, one formula, one application and one common mistake for every major topic.